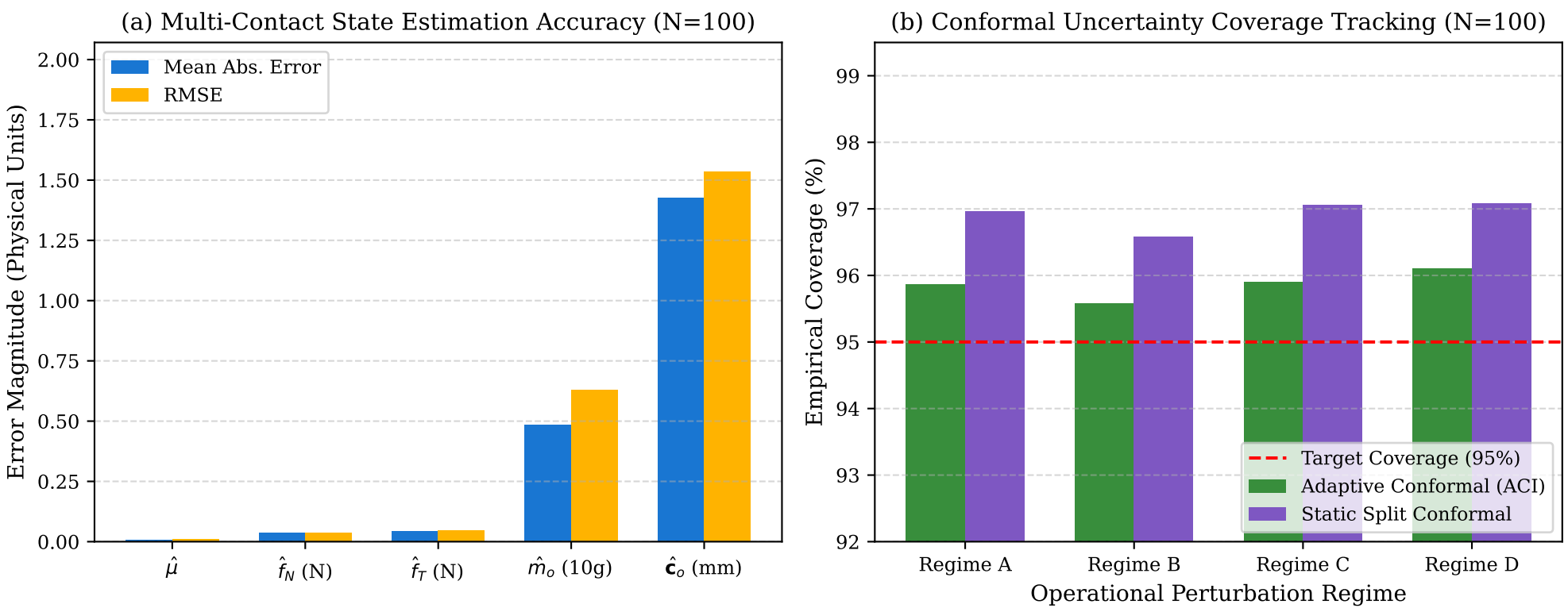
(a) Multi-Contact State Estimation Accuracy (N=100)
Mean Abs. Error
RMSE
Error Magnitude (Physical Units)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$\hat{\mu}$
$\hat{f}_N$ (N)
$\hat{f}_T$ (N)
$\hat{m}_o$ (10g)
$\hat{\mathbf{c}}_o$ (mm)
(b) Conformal Uncertainty Coverage Tracking (N=100)
Empirical Coverage (%)
99
98
97
96
95
94
93
92
Target Coverage (95%)
Adaptive Conformal (ACI)
Static Split Conformal
Regime A
Regime B
Regime C
Regime D
Operational Perturbation Regime